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DESIGN OF A SMARTART CLASSROOM IN ART HISTORY: A LEARNING EXPERIENCE WITH SELF-REGULATED SERIOUS GAMES

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Abstract

Today's society is moving towards new forms of education, both regulated and non-regulated. Learning in the 21st century uses frequently the B-Learning spaces. Those spaces allow to insert of hypermedia resources (videos, virtual laboratories ...) that make easier the personalized monitoring of the student and process-oriented feedback in real time. Additionally, recent research has made new proposals of what has been called Intelligent Tutoring Systems (ITS). These systems facilitate the development of Self-Regulated Learning (SRL) and have been very effective to increasing students' motivation and achieving effective learning. In particular, the design of virtual classrooms where serious games techniques with SRL are used resemble to be really effective. This simulation structure together with the development of interactive activities with the computer in which a process-oriented feedback is included seems to enhance the motivation and creativity of the apprentices. It has been proven that the use of this methodology explains 72% of the variance of the students' learning results. Therefore, throughout this paper we propose the design of a Virtual and Intelligent Classroom called SmartArt. This classroom can be adapted to different types of students interested in the History of Art. On the one hand, it adapts to the training needs of art history knowledges for different undergraduates, such as History and Heritage Degree, Media Communication Degree and Tourism Degree. Also, to non-regular students such as university extracurricular courses. The design of this virtual classroom is intended to be developed through integration in educational web platforms, such as Moodle, of serious games designed using 2D design implemented in the game engine Unreal Engine as a graphic adventure videogame. The different characters included in this graphic adventure explain the main milestones of the History of Art, focused in a first moment, in those corresponding to the Ancient and Medieval Ages, whose development is articulated in three different difficulty levels (Beginner, Intermediate and Advanced). The learning structure in this Virtual Classroom uses self-regulated questions about each learning milestone that is presented in the video game. It always starts from a previous knowledge analysis that is expanded with an interactive glossary, then serious games that include questions of inquiry, word games, ... are proposed. Once the student gives an answer, he receives an answer with a justification for his success or failure by process-oriented feedback (he is redirected to the correct content). Likewise, each unit includes a rubric in which the student is given adjust information on the steps to achieve an effective learning. This proposal includes used examples of visualization of the different hypermedia resources and self-regulated design.

Keywords: smartart classroom, self regulated learning, serious games, art history, virtual university, gamification.

1 INTRODUCTION

1.1 Current society and new knowledge access methodologies

Today's society is moving towards new forms of education, both regulated and non-regulated, based on the realization of training courses on various subjects, mainly carried out in B-Learning environments. Recent research considers this type of training offer a very efficient tool for the learning of citizens throughout their lives [1]. In this line, the use of Flipped Learning (FL) can be considered as a new form of access to knowledge. This type of teaching is based on understanding the teaching-learning process as a bidirectional interaction space, in which the teacher is a guide in the student's learning process, but not the center as it was in traditional environments. Margulieux, McCracken, and Catrambone [2] differentiate the following virtual learning spaces (see Table 1).

- Flipped Blend (virtual exposure to the contents and feedback on the face-to-face application - F2F-).
- Supplemental Blend (exposure to F2F contents and virtual feedback on the application of the contents).
- Replacement Blend (exposure to contents and F2F and virtual feedback on the application of the contents).

As an example, the authors offer what they call the Taxonomy of Instructional Experience (MIX taxonomy), see Table 1.

Table 1. Adaptation of the mixed instruction taxonomy of Margulieux, McCracken, and Catrambone (2016)

Transmission path of the instructor				Application of information
Receiving information	Teacher's transmission	Interaction Face to Face (F2F)	Teacher mediation	
	Hybrid readings (virtual part face-to-face part)	Blended	Practices hybrid (face-to-face part virtual part)	
	Use of technology	Combination online	Mediation of technology	
Delivery route by technology				

In the FL approach, it is necessary to design the learning environment using as many didactic elements [3] as technological elements [4].

In summary, the FL methodology implies a pedagogical redesign on what, how and when to teach and evaluate [5,6] which facilitates the adaptation of the teaching and learning processes to the new social learning environment that includes the use of new technologies. This instruction is developed both inside and outside the educational spaces throughout the life (longlife) of the apprentices [7]. Moreover, the structure of information in these virtual spaces must be made attractive to the learner and must include elements that promote the development of Self-Regulated Learning (SRL) [8]. To achieve this, it is necessary to introduce quizzes in the videos with feedback to the learner's answers [9, 10].

1.2 Intelligent Tutoring Systems and Self-Regulated Learning.

Likewise, the FL Methodology has been linked in recent years to the Intelligent Tutoring Systems (ITS) [11] (see Figure 1). This pedagogical design makes learning in simulation environments very useful for the practical application of concepts [12]

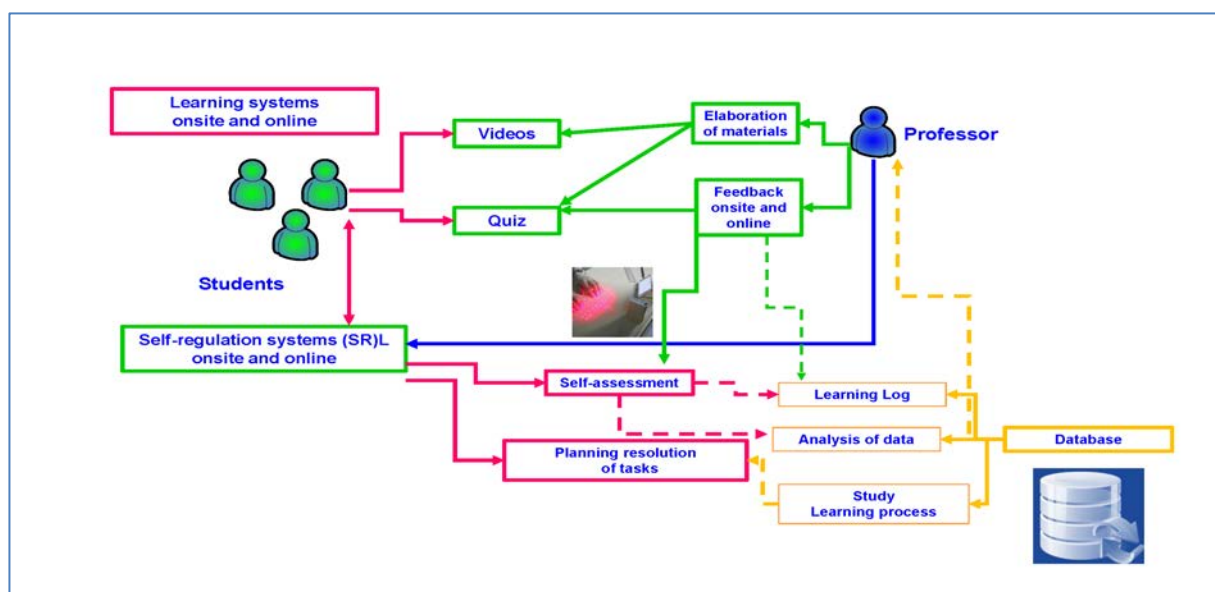


Figure 1. SRL from ITS in FL environments. Sáiz et al. 2018 p. 138 [13]

These automatic interactive systems facilitate the development of SRL [8]. All of this will increase the motivation towards learning. In addition, the SRL enhances the acquisition of significant learning [14]. Following this understanding paradigm of the learning process, the pedagogical agents, whether face-to-face or virtual, have an important role in self-regulation and in the increase of motivation in students. It is therefore important that a good didactic programming of learning objectives and evaluation procedures is carried out. Moreover, in this design the teacher must plan the type of feedback that will give to the answers of the students during the learning process. In this last aspect, virtual learning environments provide the opportunity to perform this feedback in real time [15], in what has been called MetaTutoring. It is noteworthy that students who participate in pedagogical actions in which intelligent tutoring actions are included, show a high satisfaction with this design and in addition a deeper learning achievement has been observed [16]. Also, one of the issues that currently concern researchers in the field of metacognition and SRL in digital spaces is how to provide an effective feedback that facilitates the SRL. It has been proven that process-oriented feedback is more effective than product-oriented feedback [17, 18]. However, within the first one there are also several possibilities related to the reaction time to the learning response given by the learner. All this leads to induce that the mere fact of using in the LMS mechanisms of automatic tutoring does not ensure an increase in the effectiveness of the teaching-learning process [19] if it is not accompanied by a careful design and monitoring of the execution of learning of the apprentices.

1.3 Serious games and Self-Regulated Learning.

According to this a good way to apply the SRL is the use of serious games techniques. These learning spaces are designed from very structured tasks in sequential order of difficulty. Also, they use the strategy of successive approaches to the goal from the help of an animated character e.g. an avatar that guides the user's learning through verbal help and/or execution of examples that enhance the SRL. Technology can help developing self-regulated learning processes [8] in what has been called Advanced Learning Technologies (ALTs). These techniques insert the analysis of the user learning process including registration techniques and monitoring different processes within a learning loop [14]: definition of the task, concretion of the goals and planning, use of the learning strategies, and the realization of the adjustments to the plan and the strategies throughout the process, as well as a record of the information processing analyzing the cognitive, metacognitive and motivational strategies of the apprentice during the learning process.

The tasks presented to the trainees are based on a gamified learning structure called serious games [20]. This methodology is rested on the use of digital tools that facilitate interactive learning through a simulation structure in which learning aids are introduced. These aids, as already indicated before, can come from an animated figure or avatar that is guiding the learning process. The results indicate an increase of what has been called Deep learning that includes an increase in creativity and motivation in the learner. The serious games require a delicate narrative design of the activity accompanied by a precise technical design that enhances such creativity and motivation [21].

Likewise, the learning models based on serious games facilitate the simulation with the real environment and improve the labor insertion of the apprentices. Gallego-Durán et al. [22] conducted a meta-analysis research between traditional education and gamified teaching. These authors found that variables such as age, type of task, reaction time, attentional levels and motivation, among other parameters influence learning processes. In addition, they found that gamified teaching explains 72% of the variance in learning outcomes where the use of simulators explains 62% and the tasks of quiz or puzzle games 45%. It has been found that these techniques are more effective with adults' group.

In brief, since the introduction of the concept of serious games by Clark [23] in 1970, this concept has served to revolutionize education by focusing the learning process on the personal interaction of the student with a predominantly visual and educational environment oriented towards the achievement of practical objectives (compared to traditional education based on the oral transmission of mostly theoretical concepts). This technique is based on the system of video games and seeks to enhance their positive characteristics for learning. Which have been summarized in four basic principles: experimentation or learning by doing, interaction and immediate feedback, reduce the fear of error so that through the mistake can be built meaningful learning and transfer the main role to the player [23]. In the same way the forms can use 2D and 3D design technologies. Therefore, we can conclude that the use of serious games in which virtual reality is included with the use of an avatar facilitates the SRL, allows an increase in motivation related to the ease of use of the virtual reality game and the quality of learning. In addition, this technique has been shown to be very effective in contexts of regulation and restructuring of learning [24]. Therefore, the combination of serious games in a virtual

training world is a new field of research and offers a modern and efficient alternative to traditional teaching-learning methods within Industry 4.0 [21] which will facilitate the teaching reaches more users [25].

2 METHODOLOGY

The aim of this work was to design a Virtual and Intelligent Classroom, SmartArt, adaptable to different types of students interested in learning the History of Art.

- 1 An own graphic adventure has been designed to avoid problems inherent to the development of a great computer game.
- 2 It has been united under the same storyline to perform a series of mini-games or tests to be able to work on the established objectives and competences.

3 RESULTS

3.1 Development of Smart Art

Object of learning: Classical Greek Sculpture.



3.2 Methodology of intervention

- 1 Use of the figure of two avatars that guide the user's learning. On the one hand, the figure of the teacher that regulates the learning throughout the whole teaching unit and on the other hand the Figure of Phidias that is guiding the realization of concrete actions of gamification.



2 Analysis of prior knowledge.

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HISTORY OF GREEK ART

As always, the first thing to remember is what we know about the subject

And then watch this video

ESCUPTORES GRECIA CLASICA

<https://www.youtube.com/watch?v=O3y-2Vg9RQk>

3

3 Development of interactive materials and inclusion of these in the virtual platform.

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HISTORY OF GREEK ART

And the material you'll find in the virtual Ubu platform

https://ubuvirtual.ubu.es/pluginfile.php/3777475/mod_resource/content/0/Presentaci%C3%B3n%20Tema%205%20Grecia%20Escultura.pdf

UBUvirtual Equival - Internacional [es]

Material complementario

Apuntes, artículos, textos, vídeos, webs y recursos virtuales

El taller de Fidias

Arte Griego: Textos

Textos Arte Griego 2

<https://ubuvirtual.ubu.es/course/view.php?id=1815>

4

4 Gamified task 1 of collaborative work.

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HISTORY OF GREEK ART

I hope that it will be well prepared

Now that we have all the data, I propose an activity by groups".
It is a game of clues.
Each team will receive an envelope with tracks of different types related to the various sculptors (including myself). They must be paired with the name of the corresponding artist and form a board.

The detailed details can be found on the platform in a document entitled "The workshop of Phidias"

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HISTORY OF GREEK ART

Leipo

Miron

He was an expert bronze worker, mainly in pieces of metal.

Famous for the curves that marked the lines of his works

Esopos

Known for his technique of wax casts.

5

6

5 Gamified task 2 of individual work.

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HISTORY OF GREEK ART

Ordene todas las pistas en un tablero

Mirón	Policeto	Fidias	Praxíteles	Escopas	Lisipo
	He was an expert bronzista worried about the stability in positions of march	Known for the technique of wet cloths			
Famous for the curve that marked the hips of his works					

7

6 Elaboration of the evaluation criteria of the activity.

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HISTORY OF GREEK ART

COMPETENCIAS	CRITERIOS DE EVALUACIÓN
CE03: Basic knowledge of the artistic fact and the different languages, procedures and techniques of artistic production throughout history.	- Identify the characteristics and elements of the language of classical Greek sculpture and its main representatives - It relates the characteristics of the main representatives of classical Greek sculpture with their works - Describes the characteristics and elements of the language of classical Greek sculpture and its main representatives - Defines the characteristics and elements of the language of classical Greek sculpture and its main representatives using the vocabulary of the subject - Differentiates the characteristics and elements of the language of classical Greek sculpture within the set of styles of art history
CE05: Basic knowledge of iconography	- List the main themes of classical Greek sculpture - Recognizes the main themes of classical Greek sculpture

9

7 Consolidation of the acquired knowledge: Debate in the educational Forum through a gamified activity.

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HISTORY OF GREEK ART

Before concluding the issue we will check that everything has been clear through a simple and fun task through the Platform, where you will also find the result



Escultura clásica
Completa el crucigrama



HORIZONTAL

- 3. It is the best example of interest in capturing the representation of the movement
- 5. Introduced the nude in the female figure
- 6. At the head of his workshop he was in charge of the sculptural program of the Parthenon

VERTICAL

- 1. One of his best known works is the Doriforo
- 2. His sculptures dominate the space that surrounds them
- 4. It is defined by its very marked expressions, with deep eyes, bulging foreheads and parted mouths

11

12

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Como
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Fipged Classroom en Virtual"

Soluciones subidas al foro

Escultura clásica
Complete el crucigrama

HORIZONTAL

- 3. It is the best example of interest in capturing the representation of the movement Miron
- 5. He introduced the nude in the female figure Praxiteles
- 6. In front of his workshop he was in charge of the sculptural program of the Parthenon Fidas

VERTICAL

- 1. One of his best known works is the Doriforo Policleto
- 2. His sculptures dominate the space that surrounds them Lisipo
- 4. It is defined by its very marked expressions, with deep eyes, bulging foreheads and parted mouths Escopas

13

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Como
"Fipged Learning: Diseño y Aplicación de Aprendizaje Basado en Proyectos y
Fipged Classroom en Virtual"

Once the task is done and you solve the doubts, do not forget to complete the glossary with the definitions that we have been learning, including an allusive image in each of them

Consensual Glossary

Glosario Historia del Arte Antiguo

14

- 8 Information on the evaluation criteria and weighting in the qualification of the tasks performed that are specified in the evaluation rubric. Execution is measured on a Likert scale from 1 to 5 from Excellent to Not acceptable.

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Como
"Fipged Learning: Diseño y Aplicación de Aprendizaje Basado en Proyectos y
Fipged Classroom en Virtual"

As you know from the teaching guide, the task was part of 40% of the practices and for everyone to be clear about how the practice is going to be qualified, the Evaluation Rubric is also established for each assessment criterion already set, with its corresponding percentage. You can find all this in the next pictures

ASSESSMENT CRITERIA	EXCELLENT	VERY GOOD	GOOD	LITTLE ACCEPTABLE	CLEARLY INSUFFICIENT
	5	4	3	1-2	0

15

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Como
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Fipged Classroom en Virtual"

ASSESSMENT CRITERIA CE03-10%	EXCELLENT	VERY GOOD	GOOD	LITTLE ACCEPTABLE	CLEARLY INSUFFICIENT
She or he identifies the characteristics and elements of the language (80% to 100%) of classical Greek sculpture and its main representatives	5	4	3	2-1	0
He or she relates the characteristics of the main representatives of classical Greek sculpture with their works	5	4	3	2-1	0

16

4 CONCLUSIONS

The design of the SmartArt, has been made emphasizing the rigor in the contents of History of Art, in this case of the Classical Greek Sculpture. Secondly, the methodological procedures of self-regulated learning have been used through the figure of an avatar that has been regulating the learning process from the technique of successive approximations to the goal and self-regulated questions directed to the inquiry. These questions work the metacognitive strategies of: 1) Orientation towards the task: a) the focus in the learning task, b) Analysis of previous knowledge; 2) Planning the resolution of the task: establish the steps to solve the task; 3). Monitor the completion of the task and 4) Reflection on the answers: analysis of possible errors and return to the beginning of the resolution if necessary. The use of metacognitive strategies through the figure of the avatar has been implemented both in individual learning tasks and in group learning tasks, facilitating collaborative learning in the discussion interactive forums in the virtual platform. Third, evaluation rubrics have been included. These rubrics have two functions, on the one hand, informative on the evaluation criteria and on the other hand, to guide the strategies and the steps for a successful resolution of the task. Thus, it is an instrument of both evaluation and instructional training.

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